Literature Review Notes

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Contents

1 Fair Division under Ordinal Preferences: Computing Envy-Free Allocations of Indivisible Goods

Bibliographic Information

- **Authors:** Sylvain Bouveret, Ulle Endriss, Jérôme Lang
- **Year:** 2010
- **Domain:** Fair Division, Computational Social Choice
- **DOI:** 10.3233/978-1-60750-606-5-387

Main Ideas

The paper studies the fair allocation of indivisible goods among agents when only partial ordinal preferences over individual goods are available with certainty, rather than complete preferences over bundles. Agents report strict rankings over individual goods, from which incomplete preferences over bundles are induced. The central question is whether envy-free and/or Pareto efficient allocations can be guaranteed despite this incomplete information.

Research Problem

Given:

- A finite set of indivisible goods
- A finite set of agents with strict ordinal preferences over individual goods
- Induced incomplete preferences over bundles of goods via SCI-nets

Does there exist an allocation that is possibly (necessarily) envy-free?

- **Envy-freeness:** Each agent prefers the bundle she received at least as much as any of the bundles received by others.
- **Pareto efficiency:** No other allocation makes some agents better and none worse off.
- **Possible envy-freeness (PEF):** An allocation is possibly envy-free if there exists at least one completion of the agents' preferences over bundles that makes the allocation envy-free. More precisely, the allocation is possibly envy-free if there exists a set of complete preferences that are consistent with the incomplete preferences

and can complete it to makes the allocation envy-free. For instance, if Alice prefers bundle A and Bob prefers bundle B and C, then an allocation giving A to Alice and B, C to Bob is possibly envy-free as long as Alice did not express her preference for C.

- **Necessary envy-freeness (NEF):** An allocation is necessarily envy-free if it is envy-free under all possible completions of the agents' preferences, no matter how the incomplete preferences are completed. For example, if Alice prefers bundle A to B and Bob prefers B, then the allocation "Alice gets A, Bob gets B" is necessarily envy-free since Alice will never envy Bob regardless of her preferences over other bundles.

Related Questions and Issues:

- Combinatorial explosion: The space of possible bundles grows exponentially with the number of goods, making full preference elicitation infeasible. Indeed, the full elicitation costs cognitively and can lead to uncertainty on the value of the information an agent transmits.
- Computational complexity: What are the computational complexities of checking PEF and NEF? Proof of intractability and tractable cases.
- Algorithmic characterization: Proposals for simple algorithms and hypothesis under which PEF and NEF can be guaranteed.

Hypothesis

- Finite set of indivisible goods and agents
- Strict ordinal preferences over individual goods
- Strict partial ordinal preferences over bundles induced via SCI-nets (irreflexive and transitive)
- Strict linear ordinal preferences over bundles in reality (unknown to the mechanism)

Key Words

- **Indivisible goods**
- **Fair division**
- **Pairwise dominant**
- **Incomplete Ordinal preferences**
- **Envy-freeness**
- **Pareto efficiency**

- **Possibly envy-free (PEF)**
- **Necessarily envy-free (NEF)**
- **SCI-nets**
- **Monotonicity**

Notations

- $G = \{x_1, ..., x_m\}$: Set of indivisible goods $m \geq 1$.
- N: Number of agents.
- $A = \{1, ..., n\}$: finite set of n agents $n \geq 2$.
- $\rhd_N$: linear order on G representing the common ranking of goods by all agents.
- $\succ_i^*$: preference relation of agent i over bundles she might receive.
- $\pi : A \to 2^G$: allocation function assigning to each agent a bundle of indivisible goods.
  - $\pi(i) \cap \pi(j) = \emptyset$ for all $i, j \in A$ with $i \neq j$: no good is assigned to more than one agent.
  - $\bigcup_{i \in A} \pi(i) = G$: complete allocation of all goods.
- Binary relation $\succ$ or $\rhd$: relation between goods or bundles.
- strict partial order $\succ$: binary relation that is irreflexive and transitive.
- linear order $\succ$: strict partial order that is complete, i.e $X \succ Y or Y \succ X$ whenever $X \neq Y$ for all $X, Y \in G$.
- Monotonicity: if $Y \subset X$, it implies $X \succ Y$
- Reflexive closure $\succeq$ or $\unrhd$:
  - $X \succeq Y$ if and only if $X \succ Y$ or $X = Y$.
  - For two binary relations R and R', R' refines R if $R \subseteq R'$.
- Compliant: A strict partial order $\succ$ is compliant with N, if:
  - $\succ$ is monotonic.
  - $S \cup \{x\} \succ S \cup \{y\}$ for any x, y such that $x \rhd_N y$ and $S \subseteq G \setminus \{x, y\}$.
- $\succ_N$: is the smallest strict partial order that complies with N.

Methodology

- Expression of preferences is restricted to prevent combinatorial explosion.
- Preferences over bundles are not elicited directly.
- Agents provide strict rankings over individual goods (strict linear order).
  - Instead of the actual preferences of an agent i on all bundles, we have a strict partial order $\succ_i$, which represents a partial knowledge of $\succ_i^*$ obtained by inference.
- These rankings induce a partial order over bundles. Inference rule (Brams and King, Brams et al): A set A is preferred to a set B if each preference in B not in A is pairwise dominated by a preference of A not in B.
  - For instance, for $A = \{\text{apple}, \text{banana}, \text{orange}\}$ and $B = \{\text{apple}, \text{strawberry}, \text{kiwi}\}$ we have $A \setminus B = \{\text{banana}, \text{orange}\}$ and $B \setminus A = \{\text{strawberry}, \text{kiwi}\}$. If the agent i prefer banana $\succ_i$ strawberry and orange $\succ_i$ kiwi, then each items in A, not in B pairwise dominates each items in B, not in A.
- SCI-nets provide a compact and structured representation of all compatible complete preferences.
  - An SCI-net is a special type of CI-net (conditional importance network), a compact representation with no preconditions, where comparisons are only made on individual goods (singletons).
- Envy-freeness is evaluated across all possible completions.

Key Results

1. Determining whether a possibly envy-free allocation exists is computationally easy.
2. Determining whether a necessarily envy-free allocation exists is NP-hard.
3. For two agents, the NEF problem is solvable in polynomial time.
4. Necessary envy-freeness requires strong conditions, such as distinct top-ranked goods.

Limitations

- Preferences are assumed to be strict and monotonic.
- Indifference between goods is not allowed. In this article, the authors consider strict order preferences over individual goods, i.e no equality in the order is possible.
- Only ordinal preferences are considered.
- No comparison with cardinal utility-based fairness notions.

Open Questions

- Determining the existence of an envy-freeness allocation which is also necessarily Pareto efficient.
- Extending results to nonstrict SCI-nets to allow indifferences.
- Studying possible/necessary fairness under cardinal preferences.
- Bridging ordinal and cardinal approaches.

Key Ideas for My Research

- Partial preferences are a realistic modelling assumption.
- A necessarily envy-free allocation is stable to preference incompleteness, as it remains envy-free under all possible completions of the agents' unknown preferences. This fairness property is strong for partial preferences.
- SCI-nets provide a formal framework for reasoning about incomplete preferences.

Summary

This paper provides a foundational framework (definitions, propositions, proofs) for studying fair division under incomplete ordinal preferences. It highlights the trade-off between expressive power, computational tractability, and robustness of fairness guarantees.

Dimension	**Options**
Preference model	ordinal / cardinal
Level of information	complet / partiel
Fairness criteria	envy-free / Pareto
Robustness	possible / necessary

2 Handbook of Computational Social Choice (2016) - Chapter 10: Incomplete Information and Communication in Voting

Bibliographic Information

- **Authors:** Sylvain Bouveret, Ulle Endriss, Jérôme Lang
- **Year:** 2010

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Main Ideas

Research Problem

Given:

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Question: ?

Definitions:

Related Questions and Issues:

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Hypothesis

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Key Words

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Notations

Under construction

- $A = \{a_1, ..., a_m\}$: Set of alternatives representing the space of possible choices.
- $N = \{1, ..., n\}$: Set of n agents/voters.
- $\succ_i$: (total) preference order of each voter i over alternatives in A.
  - $a_i \succ_i a_j$: voter i prefers alternative a_i to a_j.
  - Compared to a permutation σ_i of A.
  - $\sigma_i(j)$ denotes the position of alternative a_j in the ranking of voter i. Give position j.
  - $\sigma_i^{-1}(j)$ denotes the alternative ranked at position j by voter i. Give alternative a_j.
- $R(A)$: set of all preference orders over A.
- $R = \langle \succ_1, ..., \succ_n \rangle$: preference profile, i.e. a tuple of preference orders, one for each voter.
- voting rule or social choice function $f(R) \in A$: selects a winning alternative based on the preference profile R. $f(R) \in \arg\max_{a \in A} s(a, R)$.
- Scheme $s(a, R)$: score each alternative $a \in A$ given a preference profile R. Measures the quality (social welfare, etc) of an alternative given the profile.
- co-winner: any alternative a with maximum score, i.e. $s(a, R) = \max_{b \in A} s(b, R)$.
- π_i: partial preference of voter i / partial order of voter i over A / transitive closure of a consistent collection of pairwise comparisons $a_i \succ_i a_k$
- $\Pi = \langle \pi_1, ..., \pi_n \rangle$: partial profile, collection of partial votes.
- completion: any vote $\succ_i$ that extends π_i.

- $C(\pi_i)$: set of all completions of partial order π_i.
- $C(\Pi) = C(\pi_1) \times ... \times C(\pi_n)$: set of all completions of partial profile Π.
- Queries for eliciting preferences:
  - Full ranking queries: ask voter i to provide a complete ranking of all alternatives in A.
  - Set ranking queries: ask voter i to provide a ranking of a subset of alternatives $S \subseteq A$.
  - Top-k queries: ask voter i to provide her top k alternatives from $\succ_i$ for some $1 \leq k \leq |A|$. Setting $k = 1$ corresponds to plurality, $k = |A|$ corresponds to the full ranking.
  - Bottom-k queries: ask voter i to provide her bottom k alternatives from $\succ_i$ for some $1 \leq k \leq |A|$.
  - Next-best alternative queries: ask voter i to provide in sequential order her most preferred alternative among those not yet ranked.
  - Pairwise comparision queries: ask voter i to state which of two alternatives, a or a' is preferred to the other.
  - Set choice queries: ask voter i to select the most preferred alternative from a subset of alternatives $S \subseteq A$. If $|S| = |A|$, this corresponds to a full ranking query.
  - Positional queries: ask voter i to provide the alternative she ranks at a given position t.
  - Approval queries: ask voter i to approve a fixed number of alternatives or any number of alternatives. Different from top-k queries as the approval queries do not require to rank the alternatives relative to each other.
  - Veto queries: ask voter i to disapprove a fixed number of alternatives or any number of alternatives. The k-veto is equivalent to (m-k)-approval queries with $m = |A|$.
  - Split queries: ask voter i to partition a subset of alternatives $S \subseteq A$ into tow or more subsets and order them such that all alternatives in a higher-ranked subset are preferred to all alternatives in a lower-ranked subset. k-approval is a special case where two blocks of size k and m-k are created.

Methodology

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Key Results

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Limitations

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Open Questions

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Key Ideas for My Research

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Summary

Dimension	Options
Preference model	ordinal / cardinal
Level of information	complet / partiel
Fairness criteria	envy-free / Pareto
Robustness	possible / necessary

3 Survey Jérôme Lang - Collective Decision Making under Incomplete Knowledge: Possible and Necessary Solutions

Bibliographic Information

- **Authors:** Jérôme Lang
- **Year:** 2020
- **Domain:** Incomplete preferences, Social Choice, Voting, Computational Social Choice
- **DOI:** –

Main Ideas

The survey studies collective decision-making problems under incomplete information about agents' preferences. A unifying approach consists in quantifying over all completions of partial preference profiles and defining *possible* and *necessary* notions of social outcomes (winners, allocations, equilibria). This framework applies across several domains of social choice, including voting, fair division, coalition formation, and non-cooperative games, and provides a robustness-oriented view of decision-making under partial knowledge.

Research Problem

Given:

- Agents whose preferences are only partially known.
- A social choice setting (voting, fair division, tournaments, games).
- A criterion or a solution (allocation, winner, fairness, efficiency).

Question: How can collective decisions be defined and computed when agents' preferences are incomplete, and what can be said about the outcomes across all possible completions of the missing information? When can we stop the elicitation process to deduce a solution?

Definitions:

Related Questions and Issues:

- Cognitive load and exponential communication burden of expression of preference.
  - Compact representation of preferences. How can we reduce the quantity of retrieved information to find a possible and/or necessary solution?
- Computational complexity induced by the exponential number of completions.
- When incomplete information is too weak to support meaningful conclusions.

Hypothesis

- In realistic collective decision-making, assuming that all the preferences are known is untenable.
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Key Words

- **Incomplete preferences**
- **Possible winner**
- **Necessary winner**
- **Preference elicitation**
- **Voting rules**
- **Fair division**
- **Responsive extension**
- **Computational complexity**
- **Social choice**
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Notations

Possible and necessary winners in voting:

- $N = \{1, ..., n\}$: set of n agents.
- $C = \{c_1, ..., c_m\}$: set of m candidates.
- $R = (\succ_1, ..., \succ_n)$: preference profile of n linear orders over C.
- $\succ_i$: linear order of voter i over C called preference relation or vote.
- Property F: mapping from preference profile R to a subset of C (e.g Condorcet winner).
  - Voting rule F: the subset of returned candidates is always non empty (e.g plurality).
  - Resolute voting rule F: the subset of returned candidates is always a singleton.
- $\triangleright$: priority relation over candidates.
- P_i: partial (or incomplete) vote, i.e a strict partial order irreflexive and transitive over C.
- $P = (P_1, ..., P_n)$: partial profile of n incomplete votes.
- $R = (\succ_1, ..., \succ_n)$: a complete profile is a completion of partial profile P if for all i $\succ_i$ is a completion of P_i.
- Completion $succ_i$ of P_i: (complete) vote that contains/satisfies all the preferences in P_i and expresses possibly other preferences to make it complete.

Fair division problem of indivisible items:

- $N = \{1, ..., n\}$: set of n agents.
- $O = \{o_1, ..., o_m\}$: set of items.
- $\succeq_i$ over 2^O: preference relation of an agent i.
- π: allocation that maps each agent to a set of items.
- Indivisible items: $\pi(i) \cap \pi(j) = \emptyset$ for $i \neq j$.
- Envy-freeness: if for all $i, j, \pi(i) \succeq_i \pi(j)$.
- Pareto-efficieny: if there is no π' such that
  - $\pi'(i) \succeq_i \pi(i)$ for all i.
  - $\pi'(i) \succ_i \pi(i)$ for some i.
- $\triangleright$: an order over O.

- Monotonicity: if $S \subseteq T$ then $T \succeq S$.
- Weakly separable: if $o, o' \neq S$ then $S \cup \{o\} \succeq S \cup \{o'\}$ if and only if $o \rhd o'$.
- $P = \langle \rhd_1, ..., \rhd_n \rangle$ profile of rankings $\rhd_i$ over O.
- $P* = \langle \succ_1, ..., \succ_n \rangle$: profile of responsive extensions $\succ_i$ of rankings $\rhd_i$.
- $T = \langle T_1, ..., T_n \rangle$: completion of $P*$, each T_i is a linear order extending $\succ_i$.

- Possible and necessary winners in voting:
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Methodology

- Preferences are collected incrementally, leading to a sequence of partial profiles
- Formal modeling of incomplete preferences and their completions.
- After each elicitation step, the current partial profile is analyzed to determine:
  - the set of possible outcomes,
  - the set of necessary outcomes,
  - whether a decision can already be made
- If a necessary outcome exists (e.g. a necessary winner under a resolute rule), the elicitation process can be stopped.

This methodology implicitly couples:

- preference elicitation,
- reasoning under incomplete information,
- and computational evaluation of criteria.

Key Results

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Limitations

- Exponential number of completions in general. We cannot consider all the completions to know if there exist necessary solutions/winners.
- Strong incompleteness makes the problem too complex, almost all the candidates are possible winners.
- Strong dependence on modeling assumptions about missing information.
- Parallelizing the elicitation process and the calculation of possible and necessary solutions can introduce complications. If elicitation can only be performed once, it is not possible to follow this model.

Open Questions

- How to design elicitation processes guided by possible and necessary outcomes?
- How to handle richer preference models (multi-attribute, multichotomies)?
- How to combine probabilistic and logical approaches to incompleteness?

Key Ideas for My Research

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Summary

Dimension	**Options**
Preference model	ordinal / cardinal
Level of information	complet / partiel
Fairness criteria	envy-free / Pareto
Robustness	possible / necessary

4 Mathematical Social Sciences - Computational Complexity of necessary envy-freeness (2024)

Bibliographic Information

- **Authors:** Haris Aziz, Ildikó Schlotter, Toby Walsh
- **Year:** 2024
- **Domain:** Fair Division, Computational Social Choice, NEF
- **DOI:** https://doi.org/10.1016/j.mathsocsci.2023.08.002

Main Ideas

This paper investigates the computational difficulty of finding fair allocations of indivisible goods when agents' preferences are only partially known. Instead of assuming precise numerical utilities, agents provide ordinal rankings over individual items, which are then used to derive consistent preferences over bundles of items. This framework allows the authors to study the fairness notion of necessary envy-freeness under incomplete preference information. The main contribution of the paper is a complexity analysis of the problem of determining whether such an allocation exists, showing that it becomes computationally intractable with three or more agents while remaining tractable in the case of two agents.

Contribution of the Paper

- Formalization of the ExistsNEF decision problem.
- Characterization of NEF allocations via prefix inequalities.
- Proof that ExistsNEF is NP-complete for three or more agents.
- Identification of tractable cases (e.g., two agents).

Research Problem

Given:

- A finite set of agents N
- A finite set of indivisible goods I
- A collection of preferences lists for each agent L

Question: Does there exist a complete allocation of the items to the agents that satisfies the property of necessary envy-freeness?

Definitions:

- **Complete allocation:** An allocation is complete if every item is assigned to some agent. Formally, this corresponds to partitioning the set of items into $|N|$ bundles, where each bundle represents the set of items allocated to an agent (Bundles may be empty).
- **Additive valuation:** An additive valuation assigns a numerical value to each item, and measures the value of a bundle by summing the values of the items it contains. An additive valuation is consistent with the ordinal preferences of an agent if it respects the ranking of items in the preference list.
- **Envy freeness (EF):** An allocation is envy-free if no agent prefers the bundle allocated to another agent over her own bundle.
- **Necessary envy-freeness (NEF):** An allocation is necessarily envy-free if it is envy-free for all additive valuations that are consistent with the agents' ordinal preferences. In other words, whatever the way we attribute values to the items, as long as we respect the order of preferences, the allocation is envy free. Whatever the way the preferences of other agents change, the agent does not envy the others.
- **Responsive set extension:** This is a method for extending preferences over single items to preferences over bundles. Under this extension, a bundle is preferred to another if every item in the second bundle can be associated via an injection with a distinct item in the first bundle that the agent strictly prefers.
- **Envy-Free up to One Item (EF1):** An allocation satisfies EF1 if, for every pair of agents, any envy disappears after removing at most one item from the bundle of the envied agent.
- **Necessary Envy-Free up to One Item (NEF1):** An allocation satisfies NEF1 if it is EF1 for all additive valuations consistent with the ordinal preferences. In other words, even under the worst-case valuation compatible with the rankings, any envy between agents can be eliminated by removing a single item from the bundle of the envied agent.

Proposition 1: An allocation is NEF if and only if $|[L^A(1:i) \cap \pi^{-1}(A)]| \geq |[L^A(1:i) \cap \pi^{-1}(B)]|$ for every pair of distinct agents A and B, and every index $i \in [|I|]$. Intuitively, for every prefix of the preference ranking of agent A, the number of items among the first i preferred items that are allocated to A must be at least as large as the number allocated to any other agent B. In other words, an allocation is NEF for an agent A if the agent A receives more items than any other agent B among its top i preferred items for all i. If A does not envy B, then for each item that B gets and that A wishes to have, A receives an item that A prefers even more.

If this inequality holds for all values of i, the allocation is necessarily envy-free and complete. If the inequality holds only for a given prefix size i, the NEF condition is satisfied only with respect to that prefix, but not necessarily for the entire allocation. If the inequality is proved for a larger i, it does not ensure the satisfability of the inequality for a smaller i.

Related Questions and Issues:

- Sometimes, it is not possible to prevent any envy. In many situations, strict envy-freeness cannot be achieved. Therefore, weaker notions such as EF1 and NEF1 are often considered.
- The decision problem ExistsNEF is NP-complete. Consequently, researchers investigate restricted cases in which the problem becomes computationally tractable.

4.1 Personal reflection

- The formulation of the problem often gives the impression that an allocation is provided as input and the task is to verify whether it satisfies NEF. However, constructing a NEF allocation from scratch corresponds to solving the *existence* problem, which implicitly requires considering all possible allocations.
- Authors: One possible approach could be to construct a partial allocation satisfying the NEF constraints and then attempt to extend it to a complete allocation.
- In previous readings, some combinations of items appear inefficient or undesirable. For example, allocating both a plane ticket and a train ticket to the same destination on the same day to the same agent may be redundant.
- This observation suggests that items could be grouped into categories. Similarly, in recommendation systems, preferences can sometimes be inferred from similar profiles. For example, if two agents share similar preferences between action and comedy movies, and one of them prefers adventure movies, this information might help infer the preferences of the other agent.

  This motivates the idea of grouping items or preferences by categories.
- Since the problem of finding complete NEF allocations is polynomial-time solvable for two agents, one possible idea would be to analyze pairwise subproblems between agents and attempt to combine these partial solutions into a global allocation.

Hypothesis

- Constant number of agents
- Agents have strict ordinal preferences over individual items
- Indivisible goods

Key Words

- **Fairness**
- **NEF**
- **Allocation**

Notations

- N: Set of agents.
- I: Set of indivisible goods.
- $A \in N$: an agent.
- L: collection of preference lists for each agent. Similar structure as the scoring vector described in the article of Kenig and Kimelfeld.
- L^A: preference list of agent A, which is a strict linear order over the set of items I.
- π: assignment/allocation of items to agents. $\pi(i)$ returns the agent to which item i is allocated.
- $\pi^{-1}(A)$: set of items allocated to agent A.
- (N, I, L): instance of the problem.

Methodology

Get a complete NEF allocation. One possible approach is to construct the allocation progressively by considering longer prefixes of the preference lists.

1. Start with the first choice of each agent and select allocations that satisfy the NEF constraints.
2. Extend the analysis to the second choices while keeping only allocations that preserve the NEF inequality.
3. Continue by considering increasingly longer prefixes of the preference rankings.

Difficulties:

- The number of possible allocations grows exponentially. Indeed the space of allocations is huge n^m for n agents and m items.
- It is difficult to determine in advance which partial allocations can be extended into complete NEF allocations.

- No obvious filtering method allows us to keep only promising partial allocations without exploring a large portion of the search space.

This combinatorial explosion leads to the NP-completeness of the ExistsNEF problem for at least 3 agents.

Proof of NP-completeness.

- Prove the NP-completeness for the case of three agents.
- Prove the NP-completeness for a fixed number of agents superior or equal to 3.

Key Results

1. A NEF1 allocation can be obtained by allocating the most valuable items among agents using a *round-robin* procedure. In a round-robin procedure, agents take turns selecting items according to a fixed order. At each step, the current agent chooses her most preferred remaining item.
2. The computational complexity of the problem ExistsNEF is a function of the number of agents involved.

Limitations

- The article studies the baseline model with complete ordinal preferences. It remains unclear whether the results extend to richer models of partial preferences.
- The preferences induced are partial over bundles, because ordinal information does not always allow for the comparison of all bundles.

Open Questions

- Are there restricted settings in which the problem of finding NEF allocations can be solved in polynomial time?
- What is the probability that a randomly generated instance admits a necessarily envy-free allocation under common probabilistic models?

Key Ideas for My Research

- This article demonstrates the complexity of the NEF problem and lead us to consider restriction that allows polynomial time solutions.

Summary

Ordinal preferences over items
(each agent ranks individual items)

$\downarrow$

Responsive set extension
(rule used to extend item rankings to bundles)

$\downarrow$

Complete allocation
(consider an allocation π which allocate bundle of objects to each agent)

$\downarrow$

Induced preferences over bundles
(use extension rule to get partial preferences over allocated bundles)

$\downarrow$

Define envy between bundles
(agent i prefers bundle of j to their own)

$\downarrow$

Determine Necessary Envy-Freeness
(using the proposition 1 requirements. No envy over any set of items allocated to some other agent / no envy for all additive valuations consistent with the ordinal preferences)

5 Vocabulary

- **Term 1:** Definition
- **Term 2:** Definition
- **Term 3:** Definition